

Smart Parking System - Code Architecture & Flow

1. Backend Architecture

Backend follows layered architecture: Controller Layer: Handles HTTP requests and responses. Service Layer: Contains business logic. Repository Layer: Communicates with database. Entity Layer: Defines database models. Security Layer: Handles JWT and Spring Security.

2. JWT Authentication Flow

LoginController -> AuthenticationService -> JwtService Step-by-step: 1. User sends email/password. 2. AuthenticationManager validates credentials. 3. JwtService creates token. 4. Token returned to frontend. 5. Frontend stores token. 6. Axios interceptor sends Bearer token. 7. JwtAuthenticationFilter validates token.

3. User Entity Flow

User Entity: - id - name - email - password - role UserRepository: - findByEmail() CustomUserDetailsService: Loads user for Spring Security authentication.

4. Parking Flow

ParkingLocation Entity -> contains ParkingSlot entities ParkingSlot: - slotCode - status - vehicleNumber - zone Admin can update slot status dynamically.

5. Booking Flow

BookingController -> BookingService -> BookingRepository BookingService responsibilities: - validate slot - create booking - calculate amount - update slot status - cancel booking

6. Payment Flow

PaymentController -> PaymentService -> PaymentRepository PaymentService: - create payment - generate transaction ID - refund payment - update payment status

7. Frontend Flow

Pages: - LoginPage - RegisterPage - UserDashboard - AdminDashboard Services: - auth.service.ts - booking.api.ts - payment.api.ts - admin.api.ts Components: - Sidebar - Topbar - Cards - Tables

8. Axios API Flow

Axios interceptor automatically attaches JWT token. localStorage -> axios interceptor -> Authorization header -> Spring Security filter

9. React Context Flow

AuthContext: - stores logged-in user - stores JWT token - provides login/logout helpers ThemeContext: - handles dark/light mode

10. Routing Flow

React Router: - public routes - protected user routes - protected admin routes Role-based route protection: - USER -> /user/* - ADMIN -> /admin/*

11. Admin Analytics Flow

Admin analytics page collects: - booking counts - total revenue - occupancy data - payment statistics Data comes from admin analytics API endpoints.

12. Error Handling

Backend: - Spring exception handling - JWT exception handling - validation handling Frontend: - toast notifications - loading states - error cards

13. Recommended Next Features

- Real payment gateway - AI parking recommendation - WebSocket live updates - Notifications system - Email verification - Forgot password system - Docker deployment